MUKESH KUMAR DHARANI

HSE Engineer | Mining & Energy Sector

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PROFESSIONAL SUMMARY

HSE Engineer with 4.5 years of field experience across underground coal gasification (UCG) and construction projects. Sustained a 1.2 million man-hour LTI-free record, consistently exceeding safety targets in high-consequence environments. Proficient in ISO 45001, ISO 14001, HIRA, JSA, RCA/CAPA, permit-to-work (PTW) systems, gas monitoring, and emergency response planning, with a proven record of building safety culture from the ground up in Greenfield and operational settings. NEBOSH IGC-certified and currently completing an M.E. in Georesources and Geoenergy Engineering (Sustainable Mining track) at Politecnico di Torino. Available for HSE roles in the GCC, Europe, and Pakistan.

PROFESSIONAL EXPERIENCE

HSE Site Engineer ***Aug 2022 – Jul 2025***

Abbas-R Construction Company, Hyderabad, Sindh, Pakistan

- Drove a **45% reduction in workplace incidents** by designing and deploying site-specific HSE plans, risk assessments, and JSAs across high-risk construction activities in full compliance with **ISO 45001** and **ISO 14001**.
- Sustained **1.2 million man-hours LTI-free** by leading 300+ toolbox talks, safety inductions, and workforce training programmes, elevating PPE compliance from **70% to 98%** site-wide.
- Delivered **95%+ compliance scores** across all internal and external audits by maintaining rigorous HSE documentation, incident investigation records, and corrective action registers.
- Reduced near-miss incidents by **35%** through behavioural safety programmes and 20+ emergency mock drills, measurably improving site emergency response readiness.
- Cut site waste by **25%** and ensured full regulatory compliance on dust, noise, and waste disposal by implementing ISO 14001-aligned environmental controls and leading workforce sustainability campaigns.

HSE Officer ***Jan 2017 – Jun 2018***

UCG Project Thar — Government of Sindh, Coal & Energy Development Department, Thar, Sindh, Pakistan

- Established HSE policies, PTW systems, and emergency response procedures from the ground up for a Greenfield underground coal gasification (UCG) project, achieving **zero LTIs** across all high-risk commissioning and operational phases.
- Conducted comprehensive HIRA and JSA for drilling, well installation, and syngas operations, identifying **25+ critical hazard scenarios** and implementing controls that reduced task-level risk ratings by **55%**.
- Deployed continuous multi-gas monitoring systems (CO, CH_4, H_2S) across active gasification zones, enabling real-time hazard detection and reducing unplanned emergency responses by **40%** over the project lifecycle.
- Delivered safety inductions, confined space rescue drills, and gas leak response exercises to **50+ workers and contractors**, establishing a safety culture in a newly mobilised, high-consequence work environment.
- Designed and implemented a confined space entry management programme for well and borehole operations, reducing exposure time by **25%** through pre-entry atmospheric testing and standby rescue protocols.

EDUCATION

M.E. — Georesources and Geoenergy Engineering *Sep 2025 – Dec 2027 (expected)*
Politecnico di Torino, Turin, Italy | Sustainable Mining track

B.E. — Mining Engineering *Jan 2013 – Dec 2016*
Mehran University of Engineering & Technology, Jamshoro, Pakistan
CGPA: 3.63/4.00 | Merit Position Holder

CERTIFICATIONS

- NEBOSH International General Certificate in Occupational Health and Safety (IGC)
- Health, Safety, and Environmental (HSE) Engineering Specialization — Khalifa University / Coursera, Mar 2026

TECHNICAL & SOFT SKILLS

HSE Systems & Compliance: ISO 45001, ISO 14001, Permit-to-Work (PTW) Systems, Environmental Compliance, Regulatory Compliance, HSE Documentation, KPI Reporting

Risk & Incident Management: Hazard Identification and Risk Assessment (HIRA), Job Safety Analysis (JSA), Root Cause Analysis (RCA), Corrective and Preventive Action (CAPA), Incident Investigation, Safety Audits, Site Inspections

Operational Safety: Emergency Response Planning, Gas Detection, Ventilation Monitoring, Confined Space Safety, Fire Prevention Systems, Underground Mining Safety, PPE Management, Toolbox Talks, Safety Training

Mining Engineering Software: MATLAB, QGIS

Programming and Development Tools: Python, SQL, VS Code, Jupyter Notebook

Data Analytics & Visualization: Data Analytics, Machine Learning, Microsoft Office, Microsoft Excel, Microsoft Power BI

Professional: Leadership, Communication, Decision Making, Team Collaboration, Stakeholder Management, Conflict Resolution, Cross-Cultural Communication

LANGUAGES

English (Professional Proficiency); **Urdu / Sindhi** (Native); **Italian** (Intermediate, B1).